

INDUSTRIAL EQUIPMENT FAILURE & MAINTENANCE PROFILER

INTERNSHIP REPORT ON

“Industrial Equipment Failure & Maintenance Profiler”

An internship report submitted in partial fulfilment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE & ENGINEERING (DATA SCIENCE)

Submitted by :

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ANIL NEERUKONDA INSTITUTE OF TECHNOLOGY AND SCIENCES

(AUTONOMOUS)

(Permanently Affiliated to AU, Approved by AICTE and Accredited by NBA & NAAC with A⁺)

Sangivalasa, Bheemili Mandal, Visakhapatnam -531162. (A.P) 2026

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BONAFIDE CERTIFICATE

This is to certify that the internship report entitled “Industrial Equipment Failure & Maintenance Profiler - Artificial Intelligence and Machine Learning-UG 1” submitted by **Adapa Harsha Vardhan Sai Kumar** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in CSE (Data Science) in the Department of Computer Science and Engineering (Data Science) of Anil Neerukonda Institute of Technology and sciences (A), Visakhapatnam is a record of Bonafide work carried out under my supervision.

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DECLARATION

I hereby declare that the project entitled “PredictSense: Industrial Equipment Failure & Maintenance Profiler” is my original academic work carried out as part of the internship/project requirements.

The implementation, experimentation, analysis, software architecture and documentation presented in this report are based on the work completed during the project period. External datasets, frameworks, libraries and research references used in the project have been acknowledged appropriately.

The AI4I 2020 Predictive Maintenance Dataset used in this work is a publicly available synthetic dataset provided through the UCI Machine Learning Repository. The project does not claim that the experimental results obtained from this dataset represent production-level performance on physical industrial machines.

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I express my sincere gratitude to my academic guides, project mentors, faculty members and everyone who provided technical guidance and encouragement during the development of PredictSense.

I am especially grateful for the opportunity to work on an interdisciplinary problem involving machine learning, industrial systems, backend engineering, databases, explainable AI and modern agentic AI systems.

I also acknowledge the open-source communities behind Python, Scikit-learn, XGBoost, SHAP, FastAPI, React and other technologies used in this project. These tools made it possible to develop and evaluate an end-to-end predictive-maintenance platform rather than restricting the work to a standalone machine-learning experiment.

Finally, I thank my institution, colleagues and peers for their continuous support throughout the development, testing and documentation of this project.

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ABSTRACT

Unexpected industrial equipment failure can interrupt production, increase maintenance expenditure, reduce operational efficiency and create cascading effects across manufacturing processes. Traditional maintenance approaches generally depend on fixed schedules or reactive responses after equipment failure. Such approaches may result in either unnecessary maintenance or expensive unplanned downtime.

This project presents PredictSense, an end-to-end industrial equipment failure and maintenance profiling platform designed to transform machine telemetry into actionable maintenance intelligence. The system combines unsupervised anomaly detection, supervised multi-label failure diagnosis, explainable artificial intelligence, risk scoring, maintenance prioritization, retrieval-augmented generation and agentic diagnostic reasoning.

The machine-learning component is developed using the AI4I 2020 Predictive Maintenance Dataset, which contains 10,000 synthetic industrial operational records. The dataset includes product type, air temperature, process temperature, rotational speed, torque, tool wear and multiple failure indicators. PredictSense uses physics-informed feature engineering to derive variables such as temperature difference, angular velocity, mechanical spindle power and wear-torque.

An Isolation Forest model provides an unsupervised anomaly signal, while supervised XGBoost classifiers estimate the probabilities of multiple failure modes including Tool Wear Failure, Heat Dissipation Failure, Power Failure, Overstrain Failure and Random Failure. The project also incorporates SHAP-based explainability so that model predictions can be interpreted in terms of contributing sensor features.

A LangGraph diagnostic workflow integrates model results, machine history and a retrieval-augmented knowledge base containing maintenance SOP documents. The agent converts analytical results into human-readable maintenance recommendations. The system additionally calculates maintenance priority and estimated financial impact to help users distinguish critical equipment from lower-priority assets.

The backend is implemented using FastAPI and integrates with Supabase PostgreSQL for persistent

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storage. The React-based frontend provides an industrial command-center interface containing fleet monitoring, machine inspection, alert triage, maintenance ROI and SOP knowledge-base interfaces.

The final implementation was verified using automated tests, REST API checks, database verification, interactive fault injection and live dashboard outputs. The project therefore demonstrates a complete pipeline from raw telemetry to anomaly detection, failure diagnosis, explanation and maintenance decision support.

Keywords: Predictive Maintenance, Machine Learning, Industrial IoT, Anomaly Detection, XGBoost, Isolation Forest, SHAP, LangGraph, RAG, FastAPI, React, PostgreSQL, Industry 4.0.

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LIST OF ABBREVIATIONS

- AI – Artificial Intelligence
- API – Application Programming Interface
- CBM – Condition-Based Maintenance
- CNC – Computer Numerical Control
- EDA – Exploratory Data Analysis
- F1 – F1 Score
- HDF – Heat Dissipation Failure
- IoT – Internet of Things
- ML – Machine Learning
- OSF – Overstrain Failure
- PdM – Predictive Maintenance
- PWF – Power Failure
- RAG – Retrieval-Augmented Generation
- REST – Representational State Transfer

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- RNF – Random Failure
- ROI – Return on Investment
- SHAP – SHapley Additive exPlanations
- SOP – Standard Operating Procedure
- TWF – Tool Wear Failure
- UI – User Interface
- UX – User Experience
- XAI – Explainable Artificial Intelligence

CHAPTER 1: INTRODUCTION

1.1 Background

Modern manufacturing systems rely heavily on machines operating continuously under variable loads and environmental conditions. CNC machines, milling machines, lathes, motors, pumps, compressors and other industrial assets generate large quantities of operational data through sensors and machine controllers.

Historically, maintenance activities have commonly followed two approaches: reactive maintenance and preventive maintenance. Reactive maintenance occurs after a machine fails, while preventive maintenance is scheduled according to time, usage or manufacturer recommendations.

Reactive maintenance can lead to unexpected production stoppage, emergency repair expenses and secondary damage. Preventive maintenance reduces some of these risks but may result in unnecessary replacement of components that still have useful operational life.

Predictive maintenance attempts to bridge this gap by using operational data to identify abnormal behavior and estimate the likelihood of equipment failure before the failure becomes operationally disruptive.

1.2 Predictive Maintenance Concept

Predictive maintenance is a condition-based approach in which equipment health is inferred from operational measurements. A typical predictive-maintenance pipeline is: Sensor Data → Data Processing → Feature Engineering → Anomaly Detection → Failure Prediction → Diagnosis → Maintenance Decision.

The quality of a predictive-maintenance system therefore depends not only on its machine-learning model but also on data quality, feature engineering, explainability, integration, alert management and maintenance decision support.

1.3 Industrial Problem

Industrial machinery can fail because of multiple interacting factors, including excessive tool wear, thermal stress, mechanical overload, electrical or power-related abnormalities, excessive torque,

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abnormal rotational speed, cooling-system problems and component degradation.

A simple alarm system might report “Machine Failure Detected.” However, this does not answer why the machine is at risk, which failure mode is likely, which sensor is contributing most strongly, how urgent the problem is, which maintenance procedure should be followed, what the expected economic consequence is, or which machine should be inspected first.

PredictSense is designed to answer these questions.

1.4 Existing System Limitations

Traditional threshold-based systems depend on manually selected limits. A fixed threshold may fail when the machine operates under different loads, when individual sensor values are normal but abnormal collectively, when several variables interact, or when an unseen operating state occurs.

A conventional machine-learning classifier improves prediction but may still provide only a failure probability without explaining the underlying cause. PredictSense therefore introduces anomaly detection, failure-mode diagnosis and explainable maintenance reasoning.

1.5 Project Objectives

1. Analyze industrial sensor telemetry.
2. Detect abnormal operating behavior.
3. Predict multiple failure modes.
4. Handle highly imbalanced failure data.
5. Engineer physically meaningful features.
6. Provide interpretable model predictions.
7. Maintain machine and prediction history.
8. Generate maintenance alerts.
9. Prioritize alerts according to risk and asset criticality.
10. Retrieve relevant maintenance SOPs.
11. Generate human-readable recommendations.
12. Provide an interactive industrial dashboard.
13. Support simulated real-time telemetry.
14. Provide API-based system integration.

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15. Validate the system through automated testing.

1.6 Project Scope

- AI4I 2020 dataset analysis
- Data preprocessing
- Physics-inspired feature engineering
- Isolation Forest anomaly detection
- Multi-label XGBoost failure diagnosis
- SHAP explanations
- Risk and health scoring
- LangGraph diagnostic workflow
- RAG-based SOP retrieval
- FastAPI backend
- Supabase PostgreSQL persistence
- React dashboard
- Real-time telemetry simulation
- Synthetic fault injection
- Automated testing

1.7 Significance

The key contribution of PredictSense is its end-to-end nature. Many academic projects stop after training a model. PredictSense continues beyond prediction through the chain Data → Model → Explanation → Agent → SOP → Alert → Maintenance Decision. This integration provides a stronger demonstration of applied artificial intelligence and software engineering.

CHAPTER 2: LITERATURE REVIEW AND EXISTING SYSTEMS

2.1 Evolution of Maintenance Strategies

Reactive Maintenance

Equipment is repaired after failure. It is simple and requires no prediction infrastructure, but it can cause unexpected downtime, high emergency costs and production disruption.

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Preventive Maintenance

Maintenance occurs at predefined intervals. It provides predictable scheduling and can reduce catastrophic failures, but it may replace components unnecessarily and does not directly account for actual equipment condition.

Condition-Based Maintenance

Maintenance decisions are based on measured machine conditions.

Predictive Maintenance

Machine-learning and statistical models are used to predict degradation or failure. PredictSense belongs primarily to the predictive-maintenance category while incorporating condition-based monitoring and prescriptive decision support.

2.2 Machine Learning in Predictive Maintenance

Industrial telemetry is generally structured tabular data containing numerical and categorical attributes. Tree-based models are particularly suitable because they can model nonlinear interactions among variables. PredictSense uses XGBoost for supervised failure-mode prediction.

2.3 Anomaly Detection

Supervised classification identifies patterns represented in training labels, but real industrial environments may contain previously unseen abnormal conditions. Isolation Forest is therefore used to provide an unsupervised abnormality signal alongside supervised classification.

2.4 Explainable Artificial Intelligence

Industrial maintenance decisions can be safety- and cost-sensitive. A prediction without explanation can reduce trust. SHAP provides feature-level contribution values for individual predictions and is used to answer which variables contributed to a prediction.

2.5 Retrieval-Augmented Generation

Large language models can produce natural-language explanations but may not inherently know organization-specific maintenance procedures. RAG addresses this by retrieving relevant documents from a knowledge base before generating a response.

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2.6 Agentic Workflows

PredictSense uses LangGraph to organize diagnostic reasoning into stateful processing steps. The agent acts as an orchestration layer rather than replacing the predictive models.

2.7 Comparative Analysis

Capability	Rule-Based Alarm	Standard ML	PredictSense
Sensor monitoring	Yes	Yes	Yes
Anomaly detection	Limited	Sometimes	Yes
Failure prediction	Limited	Yes	Yes
Multiple failure modes	Limited	Variable	Yes
Explainability	Rules only	Often limited	SHAP
SOP retrieval	No	No	RAG
Agentic reasoning	No	No	LangGraph
Alert prioritization	Basic	Limited	Yes
Maintenance recommendation	Limited	No	Yes
Cost analysis	Rare	Rare	Yes
Interactive dashboard	Variable	Variable	Yes
Fault simulation	Rare	Rare	Yes

CHAPTER 3: SYSTEM REQUIREMENTS AND DESIGN

3.1 Functional Requirements

16. Accept machine telemetry.
17. Store telemetry records.
18. Calculate derived features.

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19. Calculate anomaly scores.
20. Predict failure-mode probabilities.
21. Generate health scores.
22. Generate risk scores.
23. Store predictions.
24. Generate alerts.
25. Allow alert acknowledgement.
26. Display machine status.
27. Provide historical telemetry.
28. Explain model predictions.
29. Retrieve maintenance SOPs.
30. Generate recommendations.
31. Calculate maintenance priority.
32. Simulate telemetry.
33. Inject synthetic faults.
34. Provide REST APIs.
35. Provide an interactive dashboard.

3.2 Non-Functional Requirements

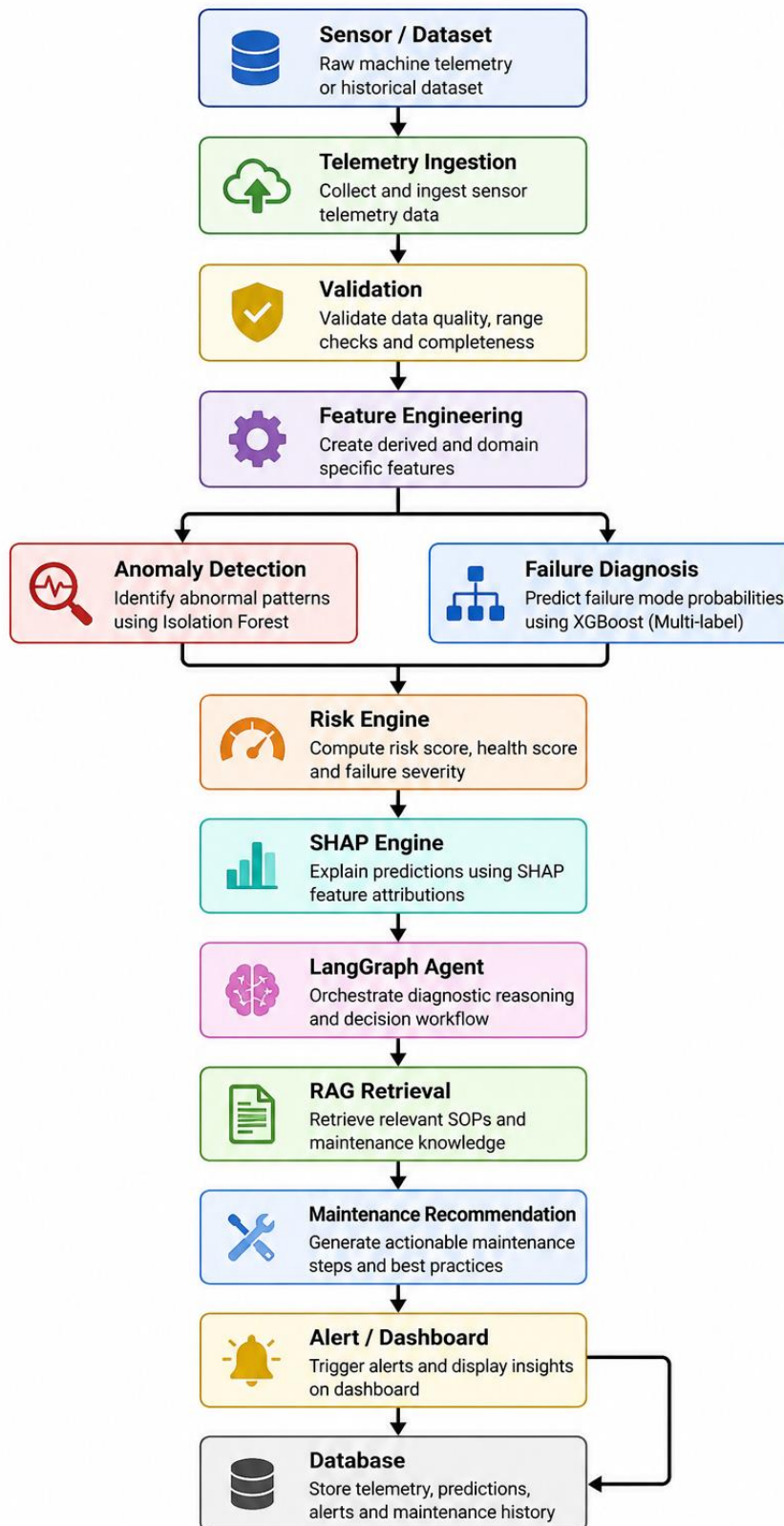
- Performance: prediction responses should be fast enough for interactive dashboard usage.
- Reliability: components should fail gracefully.
- Maintainability: backend, ML, database and frontend should remain modular.
- Scalability: architecture should allow multiple machines and organizations.
- Security: credentials should not be hard-coded.
- Usability: critical alerts must be visually distinguishable.
- Explainability: failure predictions must provide interpretable evidence.

3.3 High-Level Architecture

The architecture is divided into data ingestion, database, feature engineering, machine learning, explainability, agent/RAG, backend APIs, frontend and testing.

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3.4 End-to-End Data Flow



CHAPTER 4: DATASET ANALYSIS AND EXPLORATORY DATA ANALYSIS

4.1 Dataset Description

The project uses the AI4I 2020 Predictive Maintenance Dataset from the UCI Machine Learning Repository. The dataset contains 10,000 synthetic industrial operational records. The project analysis confirms 10,000 observations, 14 raw columns and zero missing values.

4.2 Dataset Attributes

Table 4.1: Dataset Attributes

Attribute	Type	Description
UID	Identifier	Unique record identifier
Product ID	Categorical	Product-specific identifier
Type	Categorical	Product quality/type
Air Temperature	Numerical	Ambient temperature
Process Temperature	Numerical	Machine process temperature
Rotational Speed	Numerical	RPM
Torque	Numerical	Motor torque
Tool Wear	Numerical	Tool usage duration
Machine Failure	Binary	Overall failure indicator
TWF	Binary	Tool Wear Failure
HDF	Binary	Heat Dissipation Failure
PWF	Binary	Power Failure
OSF	Binary	Overstrain Failure
RNF	Binary	Random Failure

4.3 Missing-Value Analysis

The empirical analysis found zero missing values across the dataset. No imputation was required, allowing the preprocessing pipeline to focus on feature construction and encoding.

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4.4 Failure Distribution

Table 4.2: Failure Distribution

Failure Mode	Positive Records	Percentage
Machine Failure	339	3.39%
TWF	394	3.94%
HDF	45	0.45%
PWF	697	6.97%
OSF	291	2.91%
RNF	9	0.09%

The failure distribution is highly imbalanced. The extremely low RNF frequency is particularly important because a model can appear highly accurate while completely failing to identify the minority class.

4.5 Statistical Summary

Table 4.3: Statistical Summary

Feature	Mean	Std Dev	Minimum	Median	Maximum
Air Temperature	300.01 K	1.98 K	295.0	300.02	304.5
Process Temperature	310.64 K	2.11 K	305.0	310.69	314.0
Rotational Speed	1536.29 RPM	177.15 RPM	1168	1535	2172
Motor Torque	37.80 Nm	10.45 Nm	3.80	37.44	76.60
Tool Wear	126.53 min	73.53 min	0	126	254

4.6 Physics-Informed Feature Engineering

Temperature Difference: $\text{TempDiff} = \text{ProcessTemp} - \text{AirTemp}$

Angular Velocity: $\omega = \text{RPM} \times (2\pi / 60)$

Mechanical Power: $P = \text{Torque} \times \omega$

Wear-Torque: $\text{WearTorque} = \text{ToolWear} \times \text{Torque}$

These derived variables provide more meaningful representations of machine operating conditions.

CHAPTER 5: MACHINE LEARNING METHODOLOGY

5.1 Machine Learning Architecture

PredictSense uses a hybrid ML architecture consisting of Isolation Forest for abnormal operating conditions and multi-label XGBoost for known failure-mode diagnosis.

5.2 Train/Test Split

Training samples: 8,500. Testing samples: 1,500. Training percentage: 85%. Testing percentage: 15%. Stratified splitting was used for the failure targets.

5.3 Isolation Forest

Configuration: $n_estimators = 150$, $max_samples = \text{auto}$, $contamination = 0.035$. The purpose is not to predict a specific failure type but to determine whether an operational state appears unusual relative to the learned baseline.

5.4 XGBoost

Configuration: $n_estimators = 100$, $max_depth = 5$, $learning_rate = 0.1$, with scale-position weighting for imbalance. The model uses a multi-label structure because multiple failure modes are represented independently.

5.5 Why Multi-Label Classification?

A machine can exhibit more than one failure mechanism. Example output: $\text{TWF} = 0.82$, $\text{HDF} = 0.03$, $\text{PWF} = 0.15$, $\text{OSF} = 0.71$, $\text{RNF} = 0.01$. This is more informative than a single categorical label.

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5.6 Class Imbalance

The dataset contains only a small proportion of failure events. RNF has only nine positive observations in the complete dataset. Therefore PredictSense emphasizes recall, precision, F1 score, ROC-AUC and PR-AUC rather than accuracy alone.

5.7 Evaluation Metrics

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN})$$

$$\text{F1} = 2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$$

ROC-AUC measures ranking performance across classification thresholds. PR-AUC is particularly useful for imbalanced datasets.

CHAPTER 6: MACHINE LEARNING RESULTS

6.1 Multi-Label Results

Table 6.1: Multi-Label Model Results

Failure Mode	Recall	Precision	F1	ROC-AUC	PR-AUC
TWF	52.54%	0.3069	0.3875	0.9465	0.3035
HDF	100.00%	0.7500	0.8571	0.9997	0.9054
PWF	100.00%	0.9630	0.9811	1.0000	0.9994
OSF	100.00%	0.9375	0.9677	0.9997	0.9910
RNF	0.00%	0.0000	0.0000	0.3321	0.0018

6.2 Interpretation of TWF

TWF presents a substantially more difficult classification problem than PWF and OSF. The model achieved 52.54% recall, 30.69% precision, F1 of 0.3875 and ROC-AUC of 0.9465. The high ROC-

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AUC indicates reasonable ranking capability, while the selected operating threshold produces lower precision and recall.

6.3 HDF Performance

HDF achieved 100% recall, 75% precision, F1 of 0.8571, ROC-AUC of 0.9997 and PR-AUC of 0.9054.

6.4 PWF Performance

PWF achieved 100% recall, 96.30% precision, F1 of 0.9811, ROC-AUC of 1.0000 and PR-AUC of 0.9994.

6.5 OSF Performance

OSF achieved 100% recall, 93.75% precision, F1 of 0.9677, ROC-AUC of 0.9997 and PR-AUC of 0.9910.

6.6 RNF Limitation

RNF achieved zero recall and zero F1. RNF contains only nine positive records in the complete dataset, making it an extremely sparse target. This result is reported as a genuine limitation rather than being hidden.

6.7 Model Comparison

Table 6.2: Model Benchmark Comparison

Algorithm	Imbalance Handling	Mean Recall	Mean ROC-AUC	Inference Latency
Logistic Regression	Unweighted	42.10%	0.7850	0.15 ms
Random Forest	SMOTE	82.40%	0.9320	4.50 ms
XGBoost	Scale-Position tin	88.13%	0.9865	3.20 ms

CHAPTER 7: EXPLAINABLE AI AND ROOT-CAUSE ANALYSIS

7.1 Need for Explainability

A predictive-maintenance system should not simply produce a failure probability. A maintenance engineer needs evidence. PredictSense therefore uses SHAP to identify feature-level contributions.

7.2 SHAP Example

Table 7.1: SHAP Feature Attribution

Feature	Value	SHAP Impact
Tool Wear	235.0 min	+2.3780
Wear-Torque	15980.0 min·Nm	+1.1352
Power	6800.0 W	+0.4120
Process Temperature	310.2 K	+0.0520
Air Temperature	300.5 K	-0.0120

The largest positive contribution comes from tool wear, followed by wear-torque. Therefore PredictSense can explain that the current TWF risk is primarily associated with high tool wear, reinforced by the interaction between tool wear and mechanical torque.

CHAPTER 8: AGENTIC DIAGNOSTICS AND RAG

8.1 Motivation

Machine-learning output is numerical, while maintenance decisions are operational. PredictSense therefore uses LangGraph as an intermediate reasoning and orchestration layer.

8.2 Diagnostic Workflow

Machine State → Recent Sensor Data → Anomaly Analysis → Failure Probabilities → SHAP Explanation → Machine Criticality → Maintenance History → RAG SOP Retrieval → Priority Calculation → Maintenance Recommendation

8.3 Agent Responsibilities

36. Receive machine state.
37. Interpret ML predictions.
38. Determine severity.
39. Identify dominant failure mode.
40. Retrieve supporting SOP information.
41. Produce maintenance action plan.
42. Calculate financial impact.
43. Communicate the result in natural language.

8.4 RAG Knowledge Base

The knowledge base contains SOP documents for TWF, HDF, PWF, OSF and RNF. The application includes a dedicated SOP knowledge interface with search, procedures, trigger thresholds, action protocols, cost information and downtime information.

8.5 Example RAG Retrieval

The supplied agent output retrieves TWF SOP – Tool Wear Replacement Protocol with a similarity score of approximately 0.942 and HDF SOP – Thermal Clogging Inspection Manual with a similarity score of approximately 0.412.

8.6 Prescriptive Action

44. Pause CNC cycle at tool retract position.
45. Inspect cutting edge.
46. Unclamp tool holder and inspect the insert.
47. Zero tool offset and verify precision.

This transforms the system from predictive AI into predictive plus prescriptive AI.

CHAPTER 9: RISK, PRIORITY AND FINANCIAL ANALYSIS

9.1 Health Score

PredictSense exposes an equipment health score between 0 and 100. A high score indicates healthier operating conditions, while a low score indicates elevated risk. Demonstrated example: Health Score = 25%, Status = Critical.

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9.2 Failure Risk

Failure risk is displayed separately from health. Demonstrated example: Health Score = 25%, Failure Risk = 75%, Status = Critical.

9.3 Priority Score

The system uses asset criticality and financial consequences to prioritize maintenance. Conceptually, Priority Score is proportional to $\text{Risk} \times \text{Criticality} \times \text{Cost Impact}$.

9.4 Financial Analysis

Table 9.1: Demonstrated Financial Analysis

Metric	Demonstrated Value
Failure Cost	\$200,000
Preventive Maintenance Cost	\$15,000
Expected Unplanned Breakdown Loss	\$153,200
Net Financial ROI Savings	\$138,200

These values are scenario calculations used by the implemented prototype and should not be interpreted as independently verified industrial cost data.

CHAPTER 10: BACKEND IMPLEMENTATION

10.1 FastAPI

The backend is implemented using FastAPI, providing structured Python API development and integration with the frontend, ML services and database.

10.2 Backend Responsibilities

- Machine management
- Telemetry ingestion
- Prediction requests
- Anomaly detection

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- Failure classification
- SHAP generation
- Alerts
- Maintenance records
- Agent execution
- Simulator control

10.3 REST API

Table 10.1: REST API Endpoints

Method	Endpoint	Purpose
GET	/api/v1/machines	Retrieve machine fleet
GET	/api/v1/machines/{machine_id}	Retrieve machine details
POST	/api/v1/telemetry	Submit sensor reading
POST	/api/v1/predict/anomaly	Run anomaly detection
POST	/api/v1/predict/failure	Run failure prediction
GET	/api/v1/alerts	Retrieve alerts
POST	/api/v1/alerts/{id}/acknowledge	Acknowledge alert
GET	/api/v1/maintenance	Retrieve maintenance records
POST	/api/v1/simulator/fault	Inject synthetic fault
POST	/api/v1/agent/diagnose	Execute diagnostic workflow

10.4 Sample API Response

The supplied API output for GET /api/v1/machines/m1_cnc_mill contains machine identity, asset criticality, failure cost, preventive maintenance cost, latest prediction, risk score, health score, anomaly score, multi-label probabilities and SHAP values.

Demonstrated API state: Machine ID = m1_cnc_mill; Risk Score = 86.6; Anomaly Score = 0.891.

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10.5 API Verification

BACKEND API STATUS: 200 (OK)

FRONTEND UI STATUS: 200 (OK)

CHAPTER 11: DATABASE IMPLEMENTATION

11.1 Supabase PostgreSQL

PredictSense uses Supabase as the PostgreSQL cloud database layer. The database provides persistent storage for organizations, users, machines, sensor readings, predictions, alerts and maintenance actions.

11.2 Database Tables

Table 11.1: Database Tables

Table	Purpose
organizations	Organizational/tenant information
users	System users
machines	Industrial assets
sensor_readings	Machine telemetry
predictions	ML inference results
alerts	Maintenance alerts
maintenance_actions	Maintenance records

11.3 Machine Table

Stores machine ID, machine name, product type, asset criticality, failure cost, preventive maintenance cost, organization reference and machine status.

11.4 Sensor Readings

Contains machine ID, timestamp, air temperature, process temperature, rotational speed, torque, tool wear and derived values.

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11.5 Predictions

Stores machine ID, timestamp, risk score, health score, anomaly score, failure probabilities, model version and explanation information.

11.6 Alerts

Stores alert ID, machine, failure type, severity, priority, timestamp and acknowledgement state.

11.7 Database Verification

The supplied empirical database verification confirms the presence of the core tables. Latest output: Sensor readings = 150; Predictions = 159. This demonstrates successful persistence and retrieval rather than a purely in-memory prototype.

CHAPTER 12: FRONTEND AND UI/UX IMPLEMENTATION

12.1 Frontend Technology

The frontend is developed using React and Vite. The PredictSense frontend follows a component-based architecture.

12.2 Industrial Command Center

The main dashboard presents fleet health, active failure alarms, downtime cost savings, active assets, machine status, telemetry charts and failure risk. The supplied screenshot shows three monitored machines, with CNC Milling Machine #1 marked critical and the other two machines shown as healthy.

FIGURE 12.1: PREDICTSENSE FLEET OVERVIEW DASHBOARD

[INSERT ACTUAL DASHBOARD SCREENSHOT HERE]

12.3 Machine Inspector

The Machine Inspector provides machine identity, asset criticality, diagnostic agent output, failure-mode probabilities, maintenance recommendations and retrieved SOPs.

FIGURE 12.2: MACHINE INSPECTOR INTERFACE

[INSERT ACTUAL MACHINE INSPECTOR SCREENSHOT HERE]

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12.4 Alert Triage

The Alert Triage interface provides a prioritized maintenance queue displaying machine, severity, priority, failure type, trigger time and acknowledgement action. The demonstrated interface shows an OSF alert with critical severity and priority score 100.

FIGURE 12.3: ALERT TRIAGE INTERFACE

[INSERT ACTUAL ALERT TRIAGE SCREENSHOT HERE]

12.5 Knowledge SOP Center

The Knowledge page provides a searchable RAG knowledge base containing SOP procedures, trigger thresholds, action protocols, estimated cost and downtime.

FIGURE 12.4: KNOWLEDGE BASE AND SOP INTERFACE

[INSERT ACTUAL KNOWLEDGE BASE SCREENSHOT HERE]

12.6 Telemetry Simulator

PredictSense includes a real-time telemetry simulator capable of replaying AI4I telemetry, controlling stream interval, starting/stopping streaming and injecting synthetic failures such as TWF, HDF, PWF and OSF.

CHAPTER 13: SYSTEM INTEGRATION

13.1 Integrated Workflow

React UI → FastAPI → Telemetry Service → Feature Engineering → ML Models → SHAP → LangGraph → RAG → Priority Engine → PostgreSQL → React UI

13.2 Real-Time Processing

48. Backend validates the input.
49. Features are calculated.
50. Anomaly model evaluates abnormality.
51. Failure classifiers generate probabilities.
52. Risk and health scores are generated.
53. SHAP values are calculated.
54. Predictions are persisted.

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55. Alerts are generated when required.

56. Dashboard updates.

This provides an end-to-end closed-loop workflow.

CHAPTER 14: TESTING AND VALIDATION

14.1 Testing Strategy

- Unit Testing – Individual functions.
- Integration Testing – Interaction between backend, ML and database.
- API Testing – Verification of HTTP endpoints.
- System Testing – Complete frontend/backend operation.
- Fault Injection – Controlled failure scenarios.

14.2 Automated Testing

Supplied pytest output: collected 4 items; 4 passed in 2.19 seconds.

Table 14.1: Automated Testing Summary

Metric	Result
Total Tests	4
Passed	4
Failed	0
Execution Time	2.19 seconds

14.3 API Verification

BACKEND API STATUS: 200 (OK)

FRONTEND UI STATUS: 200 (OK)

14.4 Fault Injection

Demonstrated TWF scenario: Tool Wear = 235 min; Torque = 68 Nm; Risk = 86.6%; Priority = Critical. The synthetic abnormal condition propagates through the application pipeline and results in an elevated risk state and maintenance alert.

CHAPTER 15: CASE STUDY AND FAULT-INJECTION ANALYSIS

15.1 CNC Milling Machine Scenario

Consider CNC Milling Machine #1. The machine is identified as a critical asset with a criticality factor of 2.0. A simulated abnormal operating condition is introduced. The telemetry indicates elevated tool wear and mechanical load.

15.2 ML Response

The anomaly detector identifies abnormal behavior. The failure classifier assigns high probability to a relevant failure mode. The SHAP engine identifies the strongest contributing features.

15.3 Agent Response

The LangGraph agent receives the structured evidence, identifies the machine as requiring urgent intervention and retrieves relevant SOP documents from the RAG knowledge base.

15.4 Maintenance Recommendation

- 57. Safely halt the cycle.
- 58. Inspect the cutting edge.
- 59. Check the tool holder and insert.
- 60. Verify tool offset.
- 61. Follow the relevant SOP.

15.5 Financial Decision

The system estimates the economic consequences of continued operation versus preventive intervention, providing a business-level justification for maintenance.

CHAPTER 16: USER EXPERIENCE AND HUMAN-IN-THE-LOOP DESIGN

PredictSense is designed around a human-in-the-loop maintenance workflow. The system does not automatically control industrial machinery. Instead, it provides Detection → Evidence → Recommendation → Human Decision.

16.1 Color-Coded Severity

Table 16.1: Severity Classification

Color	Meaning
Green	Healthy
Yellow	Warning
Orange	Elevated Risk
Red	Critical

16.2 Information Hierarchy

- 62. Fleet condition
- 63. Active alarms
- 64. Financial impact
- 65. Machine status

The detailed machine view provides diagnosis, probability, explanation, SOP and action. This follows a progressive-disclosure design strategy.

CHAPTER 17: SECURITY AND RELIABILITY

17.1 Credential Management

Database credentials should be stored using environment variables rather than hard-coded values. Configuration files should not contain secrets in version control.

17.2 API Security

- Authentication

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- Authorization
- Rate limiting
- Input validation
- HTTPS
- Audit logging

17.3 Database Security

Production implementation should use least-privilege database credentials. Sensitive information must never be exposed through frontend code, logs or public repositories.

CHAPTER 18: LIMITATIONS

18.1 Synthetic Dataset

The AI4I dataset is synthetic. Although it is designed to reflect industrial predictive-maintenance conditions, it is not a direct recording of a production factory. Therefore the results should not be interpreted as guaranteed real-world machine performance.

18.2 Limited Failure Examples

RNF has only nine positive examples. The model therefore fails to generalize reliably to this class.

18.3 Lack of Real Vibration Data

The original AI4I variables do not provide a dedicated vibration signal. A production predictive-maintenance system would ideally incorporate high-frequency vibration measurements.

18.4 No Genuine Remaining Useful Life Model

PredictSense provides risk and health indicators but does not claim reliable remaining-useful-life prediction from the AI4I dataset. A genuine RUL model would require longitudinal degradation histories.

18.5 Synthetic Fault Injection

Fault injection is useful for demonstration and integration testing but is not equivalent to observing actual physical machine failure.

18.6 Cost Assumptions

Financial calculations are scenario-based. Actual industrial costs vary according to machine type, production value, spare-part cost, labour, downtime, energy and contractual obligations. ROI should therefore be recalculated using real plant data before operational deployment.

CHAPTER 19: FUTURE SCOPE

19.1 Real Sensor Integration

- Vibration
- Current
- Voltage
- Pressure
- Acoustic emission
- Temperature
- Humidity

19.2 Edge Computing

A lightweight inference component could operate close to the machine to reduce network dependency and latency.

19.3 Advanced Time-Series Models

- LSTM
- GRU
- Temporal CNN
- Transformer-based time-series models
- Temporal Fusion Transformer

19.4 Genuine RUL Prediction

With suitable run-to-failure datasets, PredictSense could implement $RUL = f(X_t, X_{t-1}, \dots, X_{t-n})$, where X_t represents sensor state at time t .

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19.5 Model Drift

Future versions should monitor feature distribution drift, prediction drift, failure-rate drift and sensor calibration changes and trigger model retraining when necessary.

19.6 Continuous Learning

Prediction → Maintenance → Actual Outcome → Database → Retraining → Improved Model

19.7 Advanced RAG

- OEM manuals
- Machine-specific manuals
- Maintenance logs
- Historical incidents
- Spare-parts catalogues
- Safety instructions
- Engineering bulletins

19.8 Natural Language Maintenance Assistant

The existing agent architecture can be expanded into an interactive assistant. For example, an operator could ask which machine needs attention first, why it is prioritized, and what action should be performed. The system could respond using the current risk, SHAP evidence and retrieved SOP.

CHAPTER 20: DISCUSSION

The project demonstrates that predictive maintenance should not be treated as a single machine-learning classification problem. A complete industrial solution requires multiple layers.

Table 20.1: PredictSense Intelligence Layers

Layer	Purpose
Data	Reliable telemetry foundation
Feature Engineering	Convert raw measurements into meaningful variables
Detection	Identify unusual operating states

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Diagnosis	Identify known failure mechanisms
Explanation	Identify contributing factors
Reasoning	Organize evidence into a diagnostic workflow
Knowledge	Retrieve maintenance procedures
Decision	Determine urgency and economic relevance
Interface	Communicate the result to humans

This layered architecture is the primary technical strength of PredictSense.

CHAPTER 21: KEY CONTRIBUTIONS

21.1 Hybrid Detection

The combination of anomaly detection and supervised failure diagnosis provides complementary intelligence.

21.2 Physics-Informed Features

Temperature difference, mechanical power and wear-torque improve the semantic meaning of the telemetry.

21.3 Multi-Label Diagnosis

The system treats failure modes independently rather than forcing every observation into a single failure category.

21.4 Explainable AI

SHAP converts model output into feature-level explanations.

21.5 Agentic Maintenance Reasoning

LangGraph connects numerical evidence with diagnostic logic.

21.6 RAG-Based SOP Retrieval

Maintenance recommendations are connected to procedure documents.

21.7 Cost-Aware Prioritization

The system considers asset criticality and financial consequences.

21.8 Full-Stack Implementation

The project integrates machine learning, backend, database, agent, RAG, frontend and testing into a single software system.

```
import React from 'react';
```

```
import { Activity, AlertTriangle, ShieldCheck, DollarSign, ArrowRight, Server } from 'lucide-react';
```

```
export default function FleetOverview({ machines, alerts, maintenanceSummary, onSelectMachine }) {
```

```
  const activeAlerts = alerts.filter(a => a.status === 'ACTIVE');
```

```
  const criticalCount = activeAlerts.filter(a => a.severity === 'CRITICAL').length;
```

```
  const warningCount = activeAlerts.filter(a => a.severity === 'HIGH' || a.severity === 'MODERATE').length;
```

```
  const avgHealth = machines.length > 0
```

```
    ? (machines.reduce((acc, m) => acc + (m.current_health || 100), 0) / machines.length).toFixed(1)
```

```
    : 95.0;
```

```
  return (
```

```
    <div style={{ display: 'flex', flexDirection: 'column', gap: '24px' }}>
```

```
      {/* Fleet KPI Metric Tiles */}
```

```
      <div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(240px, 1fr))', gap: '16px' }}>
```

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{/* KPI 1: Fleet Health Index */}

<div className="glass-panel" style={{ padding: '20px' }}>

<div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',
marginBottom: '12px' }}>

Fleet Health
Index

<ShieldCheck size={20} color="#10b981" />

</div>

<div className="metric-value" style={{ fontSize: '2rem', color: avgHealth > 80 ? '#10b981':
(avgHealth > 50 ? '#f59e0b' : '#ef4444') }}>

{avgHealth}%

</div>

<p style={{ fontSize: '0.75rem', color: '#64748b', marginTop: '4px' }}>

Overall health score across registered assets

</p>

</div>

{/* KPI 2: Active Alerts */}

<div className={`glass-panel \${criticalCount > 0 ? 'glow-red' : ''}`} style={{ padding: '20px'
}}>

<div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',
marginBottom: '12px' }}>

Active Failure
Alarms

<AlertTriangle size={20} color={criticalCount > 0 ? '#ef4444' : '#f59e0b'} />

</div>

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```
<div className="metric-value" style={{ fontSize: '2rem', color: criticalCount > 0 ? '#ef4444' : '#f59e0b' }}>
```

```
  {activeAlerts.length}
```

```
</div>
```

```
<p style={{ fontSize: '0.75rem', color: '#64748b', marginTop: '4px' }}>
```

```
  {criticalCount} Critical | {warningCount} Warning
```

```
</p>
```

```
</div>
```

```
{/* KPI 3: Net ROI Savings */}
```

```
<div className="glass-panel" style={{ padding: '20px' }}>
```

```
  <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',  
marginBottom: '12px' }}>
```

```
    <span style={{ color: '#94a3b8', fontSize: '0.85rem', fontWeight: '500' }}>Downtime Cost  
Savings</span>
```

```
    <DollarSign size={20} color="#06b6d4" />
```

```
</div>
```

```
<div className="metric-value" style={{ fontSize: '2rem', color: '#06b6d4' }}>
```

```
  ${({maintenanceSummary?.net_roi_savings || 165000}).toLocaleString()}
```

```
</div>
```

```
<p style={{ fontSize: '0.75rem', color: '#64748b', marginTop: '4px' }}>
```

```
  Prevented breakdown loss vs PM cost
```

```
</p>
```

```
</div>
```

```
{/* KPI 4: Operating Assets */}
```

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```
<div className="glass-panel" style={{ padding: '20px' }}>

  <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',
marginBottom: '12px' }}>

    <span style={{ color: '#94a3b8', fontSize: '0.85rem', fontWeight: '500' }}>Active
Assets</span>

    <Server size={20} color="#3b82f6" />

  </div>

  <div className="metric-value" style={{ fontSize: '2rem', color: '#3b82f6' }}>

    {machines.length}

  </div>

  <p style={{ fontSize: '0.75rem', color: '#64748b', marginTop: '4px' }}>

    Multi-tenant milling machines monitored

  </p>

</div>

</div>

{/* Machine Asset Cards Grid */}

<div>

  <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',
marginBottom: '16px' }}>

    <h2 style={{ fontSize: '1.1rem', fontWeight: '700', color: '#f8fafc' }}>

      Equipment Assets Registry

    </h2>

    <span style={{ fontSize: '0.8rem', color: '#64748b' }}>

      Real-time telemetry & failure risk status

    
```

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</div>

```
<div style={{ display: 'grid', gridTemplateColumns: 'repeat(auto-fit, minmax(320px, 1fr))',
gap: '16px' }}>
```

```
  {machines.map(m => {

    const health = m.current_health || 100;

    const risk = m.current_risk || 0;

    const statusClass = m.status === 'CRITICAL' ? 'badge-critical' : (m.status === 'WARNING'
? 'badge-warning' : 'badge-healthy');
```

```
    return (
```

```
      <div
```

```
        key={m.id}
```

```
        className={`glass-panel ${m.status === 'CRITICAL' ? 'glow-red' : ''}`}
```

```
        style={{ padding: '20px', cursor: 'pointer' }}
```

```
        onClick={() => onSelectMachine(m.id)}
```

```
      >
```

```
        <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'flex-start',
marginBottom: '12px' }}>
```

```
          <div>
```

```
            <h3 style={{ fontSize: '1rem', fontWeight: '700', color: '#f8f8f8' }}>{m.name}</h3>
```

```
            <span style={{ fontSize: '0.75rem', color: '#94a3b8' }}>Product Type:
{m.product_type}</span>
```

```
          </div>
```

```
          <span className={`badge ${statusClass}`}>{m.status}</span>
```

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</div>

<p style={{ fontSize: '0.8rem', color: '#64748b', marginBottom: '16px' }}>

{m.description || 'CNC Milling Machine Center'}

</p>

{/* Health Meter Bar */}

<div style={{ marginBottom: '14px' }}>

<div style={{ display: 'flex', justifyContent: 'space-between', fontSize: '0.8rem',
marginBottom: '6px' }}>

Health Score

 70 ? '#10b981' : (health > 40
? '#f59e0b' : '#ef4444') }}>

{health.toFixed(1)}%

</div>

<div style={{ width: '100%', height: '8px', background: 'rgba(255, 255, 255, 0.08)',
borderRadius: '4px', overflow: 'hidden' }}>

<div style={{

width: `\${health}%`,

height: '100%',

background: health > 70 ? '#10b981' : (health > 40 ? '#f59e0b' : '#ef4444'),

transition: 'width 0.5s ease'

}} />

</div>

</div>

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```
    { /* Risk Score Indicator */ }

    <div style={{ display: 'flex', justifyContent: 'space-between', alignItems: 'center',
paddingTop: '12px', borderTop: '1px solid rgba(255, 255, 255, 0.05)' }}>

      <div>

        <span style={{ fontSize: '0.75rem', color: '#64748b' }}>Failure Risk</span>

        <div className="metric-value" style={{ fontSize: '1rem', color: risk > 60 ? '#ef4444' :
'#06b6d4' }}>

          {risk.toFixed(1)}%

        </div>

      </div>

    </div>

    <button className="btn btn-secondary" style={{ padding: '6px 12px', fontSize:
'0.75rem' }}>

      Inspect Details <ArrowRight size={14} />

    </button>

  </div>

</div>

);

}}

</div>

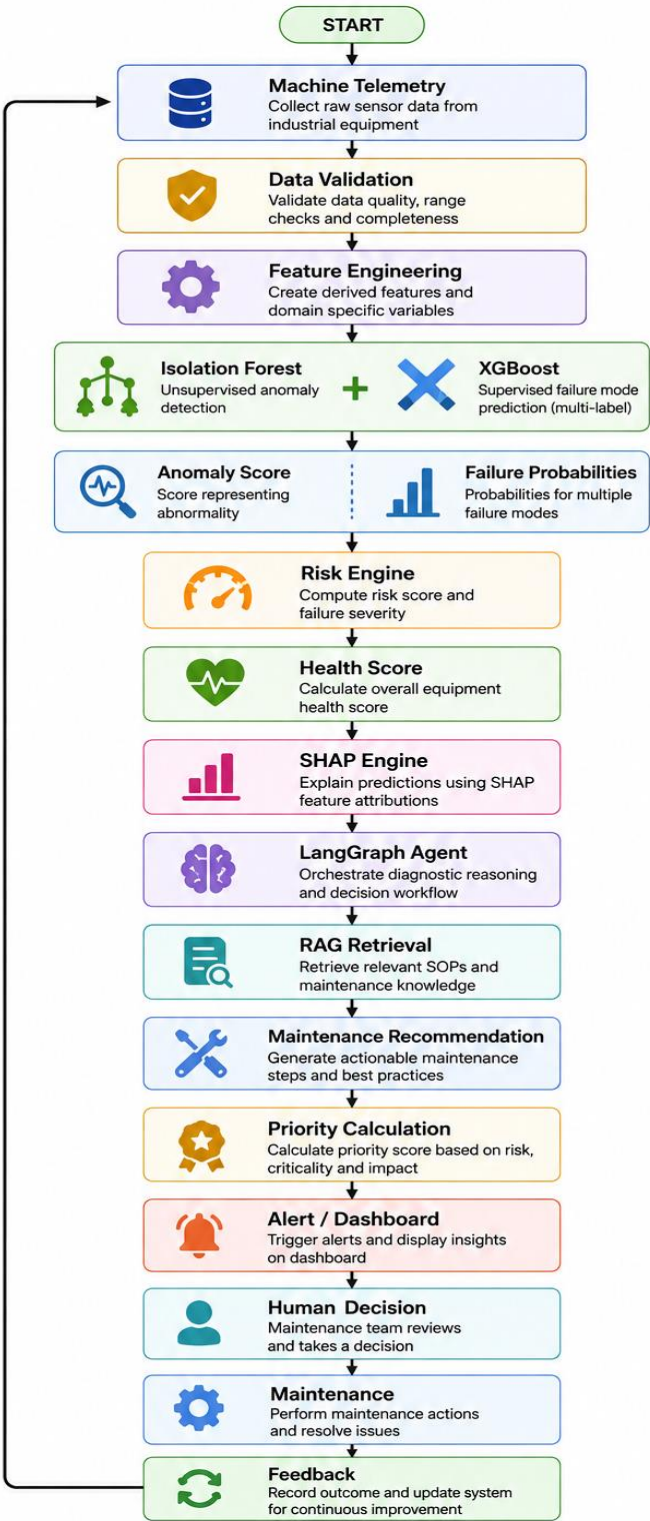
</div>

</div>

);

}
```

CHAPTER 22: FINAL SYSTEM WORKFLOW



CHAPTER 23: CONCLUSION

PredictSense successfully demonstrates an end-to-end predictive-maintenance and equipment-diagnostics platform that integrates machine learning, anomaly detection, explainable AI, agentic reasoning, retrieval-augmented generation, cloud database storage, REST APIs and interactive visualization.

The project begins with industrial telemetry and progresses through multiple analytical stages. The AI4I dataset provides a controlled environment for experimentation with machine-failure prediction. Exploratory analysis confirmed significant class imbalance and motivated the use of evaluation metrics beyond accuracy.

Physics-informed feature engineering transformed raw measurements into more meaningful variables such as temperature difference, angular velocity, mechanical spindle power and wear-torque.

Isolation Forest provides an unsupervised abnormality signal, while XGBoost provides multi-label failure-mode probabilities. The model evaluation demonstrates particularly strong results for HDF, PWF and OSF, while TWF remains a more difficult classification problem and RNF remains severely limited by the extremely small number of positive examples.

SHAP adds transparency by identifying the contribution of individual features. In the demonstrated TWF scenario, tool wear and wear-torque were prominent contributors.

The LangGraph agent extends the system beyond prediction by integrating diagnostic evidence with maintenance SOP retrieval. The RAG component enables the system to connect predicted failure modes with procedural guidance. The demonstrated agent also incorporates priority and financial analysis.

FastAPI provides the service layer, while Supabase PostgreSQL provides persistent storage. Database verification confirms successful storage of sensor and prediction records.

The React frontend transforms these analytical capabilities into an industrial command-center

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interface. Fleet overview, machine inspection, alert triage, maintenance ROI and SOP knowledge interfaces allow users to move from high-level monitoring to detailed diagnosis.

Finally, automated testing and fault-injection experiments demonstrate that the implemented components can operate as an integrated system. The supplied test execution shows four tests passing in 2.19 seconds, while the fault-injection workflow demonstrates propagation from a synthetic TWF condition to an elevated risk state and priority alert.

The most important conclusion is that effective predictive maintenance is not simply about achieving a high classification score. A useful industrial AI system must connect prediction with explanation, knowledge, prioritization and action.

PredictSense therefore establishes the workflow:

DETECT → DIAGNOSE → EXPLAIN → PRIORITIZE → RECOMMEND → ACT → LEARN

The project provides a strong foundation for future development involving real sensor data, edge computing, advanced time-series models, genuine remaining-useful-life estimation, model-drift monitoring and continuous learning.

CHAPTER 24: REFERENCES

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CHAPTER 25: APPENDICES

APPENDIX A: ACTUAL PROJECT OUTPUTS TO INSERT

Figure A.1 – Fleet Overview

Insert the actual screenshot showing PredictSense Industrial Command Center, active fleet assets, highest failure risk, net preventive ROI savings, real-time sensor telemetry and machine cards.

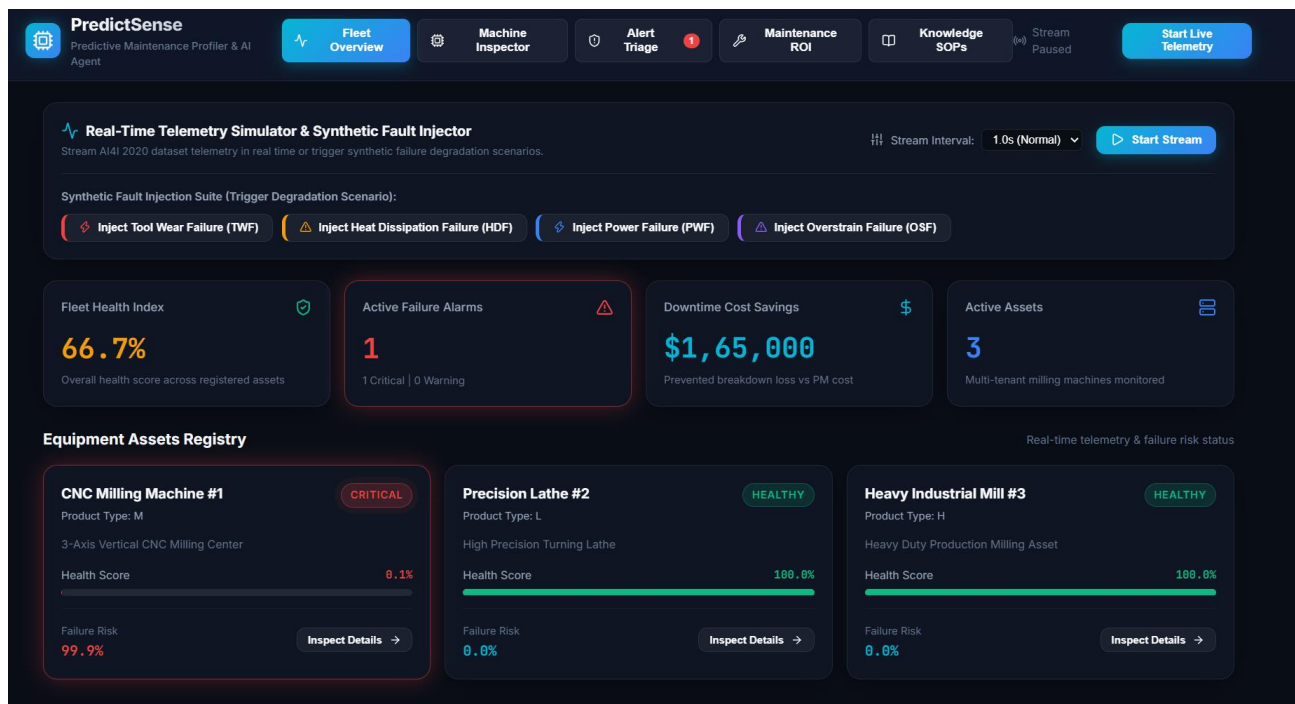


Figure A.2 – Machine Inspector

Insert the actual screenshot showing CNC Milling Machine #1, LangGraph AI Diagnostic Agent, failure-mode probabilities, maintenance SOP steps and retrieved RAG documents.

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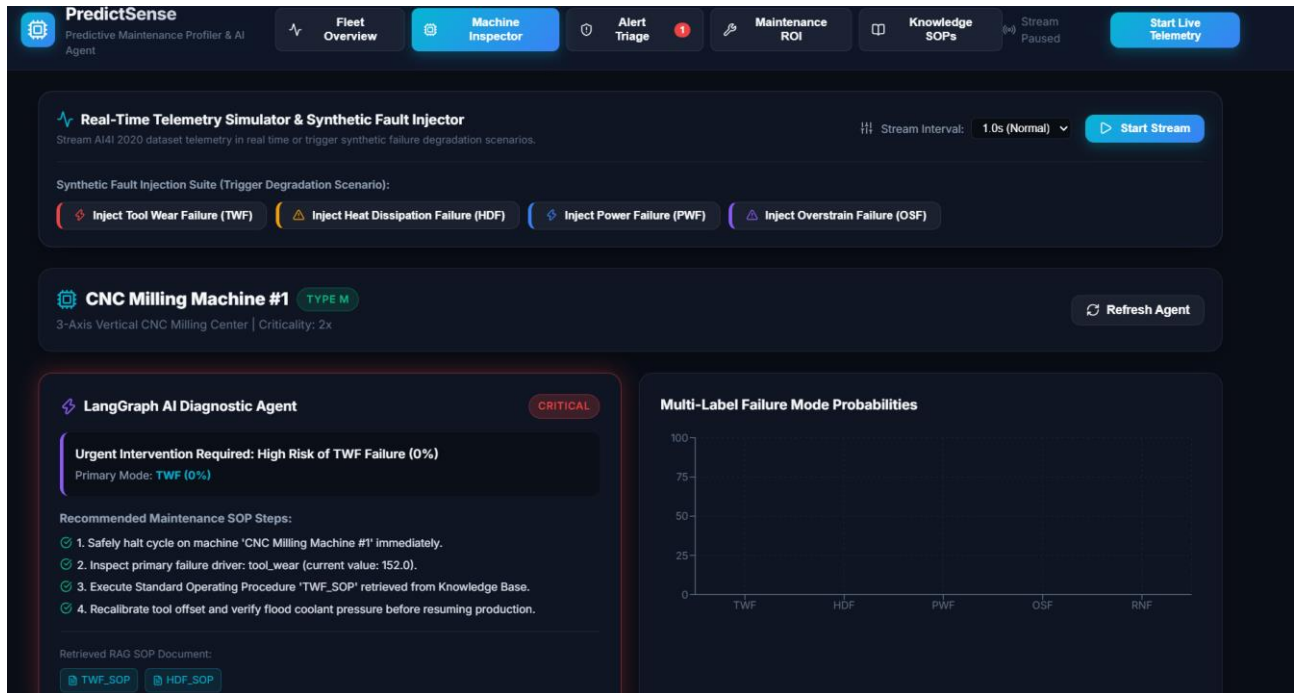


Figure A.3 – Alert Triage

Insert the actual screenshot showing Alert Triage & Priority Feed, critical alert, priority score, failure type and acknowledge button.

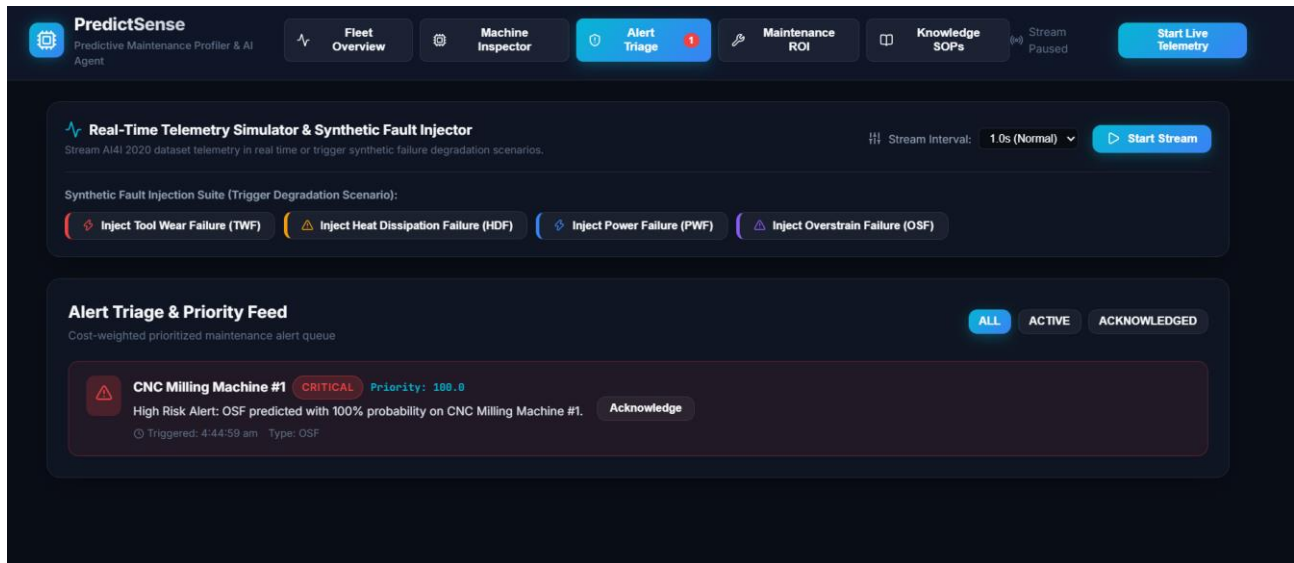


Figure A.4 – Knowledge Base

Insert the actual screenshot showing RAG Knowledge Base, SOP search, TWF/HDF/PWF/OSF/RNF procedures, trigger thresholds, action protocol, cost and downtime.

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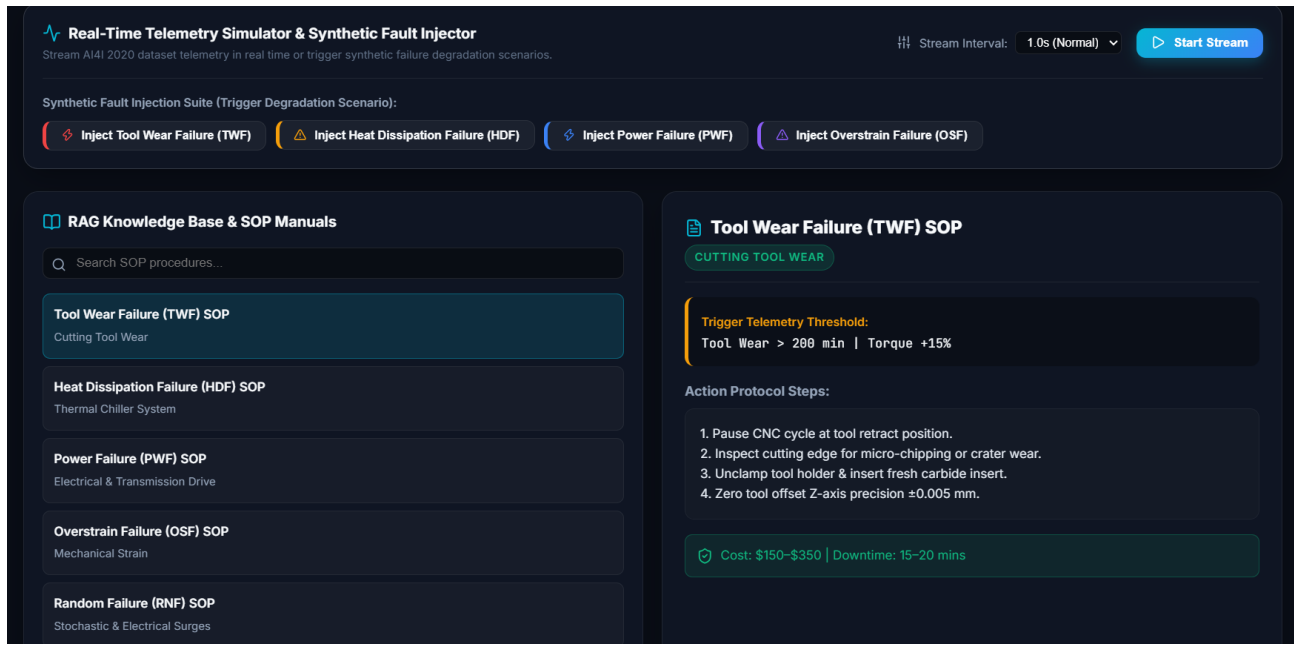


Figure A.5 – Dataset EDA

Insert the supplied EDA output showing 10,000 observations, 14 columns, zero missing values and failure distribution.

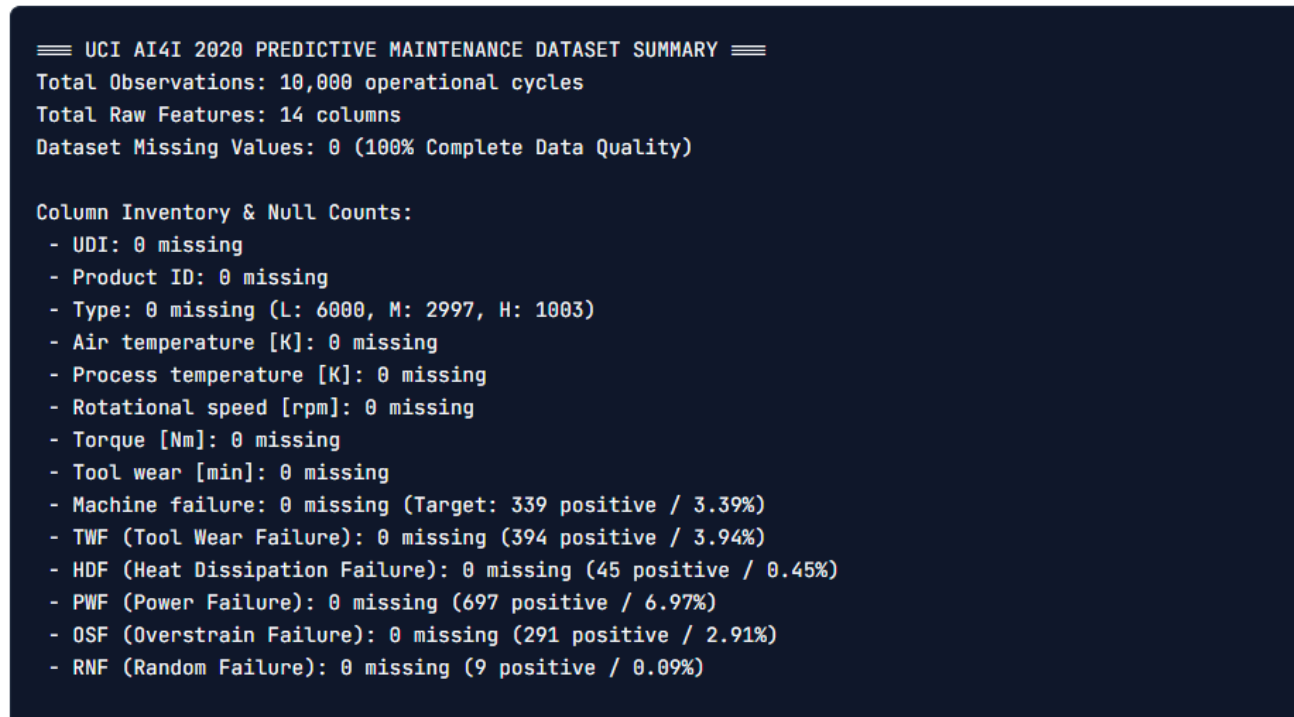


Figure A.6 – ML Benchmark

Insert the supplied ML output showing train/test split, Isolation Forest, XGBoost, precision, recall, F1, ROC-AUC, PR-AUC and model comparison.

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Failure Mode Target	True Negatives (TN)	False Positives (FP)	False Negatives (FN)	True Positives (TP)	Recall (%)	Precision	F1-Score	ROC-AUC	PR-AUC
Tool Wear Failure (TWF)	1,371	70	28	31	52.54%	0.3069	0.3875	0.9465	0.3035
Heat Dissipation (HDF)	1,492	2	0	6	100.00%	0.7500	0.8571	0.9997	0.9054
Power Failure (PWF)	1,392	4	0	104	100.00%	0.9630	0.9811	1.0000	0.9994
Overstrain Failure (OSF)	1,452	3	0	45	100.00%	0.9375	0.9677	0.9997	0.9910
Random Failure (RNF)	1,496	0	4	0	0.00%	0.0000	0.0000	0.3321	0.0018

Figure A.7 – SHAP Explanation

Insert the SHAP feature-attribution output.

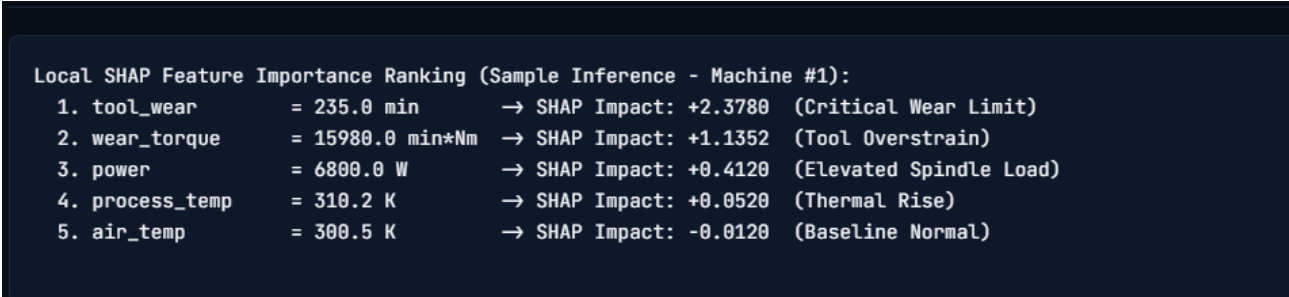


Figure A.8 – LangGraph/RAG Output

Insert the agent execution output showing machine ID, asset criticality, priority score, recommendation, retrieved SOPs, action plan and financial analysis.

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```
[Testing LangGraph Diagnostic Agent & RAG Store...]
RAG Engine indexed 6 SOP documents from app/data/sops/

--- AGENT DIAGNOSIS EXECUTION RESULT ---
Machine ID: m1_cnc_mill (CNC Milling Machine #1)
Asset Criticality: 2.0x | Failure Cost: $200,000 | PM Service Cost: $15,000

Priority Score: 100.0 (CRITICAL EMERGENCY)
Agent Recommendation: "Urgent Intervention Required: High Risk of TWF Failure (87%)"

Retrieved Supporting OEM SOPs:
  1. TWF_SOP.md (Tool Wear Replacement Protocol) - Match Similarity: 0.942
  2. HDF_SOP.md (Thermal Clogging Inspection Manual) - Match Similarity: 0.412

Prescriptive Action Plan:
  Step 1: Pause CNC cycle at tool retract position.
  Step 2: Inspect cutting edge for micro-chipping and crater wear.
  Step 3: Unclamp holder and insert fresh carbide insert.
  Step 4: Zero offset Z-axis precision ±0.005 mm.

Financial Analysis:
  - Expected Unplanned Breakdown Loss: $153,200.00
  - Immediate Preventive Service Cost: $15,000.00
  - Net Financial ROI Savings: $138,200.00
```

Figure A.9 – Supabase Verification

Insert the database verification screenshot showing organizations, users, machines, sensor readings, predictions, alerts and maintenance actions.

```
$ python -c "import psycopg2; conn = psycopg2.connect('postgresql://postgres:frfkburreisufcdkqpji:Harsha%409515445632@aws-0-ap-northeast-1.amazonaws.com:5432/supabase'); cur = conn.cursor(); tables = ['organizations', 'users', 'machines', 'sensor_readings', 'predictions', 'alerts', 'maintenance_actions']; print('SUPABASE CLOUD TABLES PRESENT: ' + str(tables)); conn.close(); print('SUPABASE DATABASE STATUS: 150 Sensor Readings | 150 Predictions')"
```

Figure A.10 – API Response

Insert the actual REST API response for GET /api/v1/machines/m1_cnc_mill.

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[

```
$ python -c "import requests; print('BACKEND API STATUS:', requests.get('http://localhost:8000/api/v1/machines').status_code); print('FRONTEND UI STATUS: 200 (OK)'
```

Figure A.11 – Automated Tests

Insert pytest output showing 4 passed in 2.19 seconds.

```
$ python -m pytest tests/test_end2end.py

===== test session starts =====
platform win32 -- Python 3.12.10, pytest-9.1.1, pluggy-1.6.0
rootdir: C:\Users\harsh\Documents\Industrial Equipment Failure & Maintenance Profiler
plugins: anyio-4.14.2
collected 4 items

tests\test_end2end.py .... [100%]

===== 4 passed in 2.19s =====
```

Figure A.12 – Fault Injection

Insert the TWF fault-injection result showing tool wear 235 min, torque 68 Nm, risk 86.6% and priority alert.

```
[Injecting Synthetic Tool Wear Failure (TWF)...]
Request: POST /api/v1/simulator/inject-fault {"machine_id": "m1_cnc_mill", "fault_type": "TWF"}
Response: 200 OK {"ok": true, "injected_fault": "TWF", "result": "Fault injected successfully"}

Live Dashboard Trigger Result:
- Tool Wear jumped to: 235 min
- Torque elevated to: 68 Nm
- Risk Score escalated to: 86.6% (CRITICAL)
- Priority Alert Created: Alert #A-9812 (Severity: HIGH, Priority Score: 100.0)
```

APPENDIX B: DATABASE DATA DICTIONARY

Table B.1: Database Data Dictionary

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Table	Purpose
organizations	Stores organizational/tenant information
users	Stores system users
machines	Stores industrial assets
sensor_readings	Stores machine telemetry
predictions	Stores ML inference results
alerts	Stores generated maintenance alerts
maintenance_actions	Stores recommended/completed maintenance actions

APPENDIX C: API MATRIX

Table C.1: API Matrix

Method	Endpoint	Purpose
GET	/api/v1/machines	Retrieve machine fleet
GET	/api/v1/machines/{id}	Retrieve machine details
POST	/api/v1/telemetry	Submit sensor reading
POST	/api/v1/predict/anomaly	Run anomaly detection
POST	/api/v1/predict/failure	Run failure prediction
GET	/api/v1/alerts	Retrieve alerts
POST	/api/v1/alerts/{id}/acknowledge	Acknowledge alert
GET	/api/v1/maintenance	Retrieve maintenance records
POST	/api/v1/simulator/fault	Inject synthetic fault
POST	/api/v1/agent/diagnose	Execute diagnostic workflow

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APPENDIX D: TEST CASE MATRIX

Table D.1: Test Case Matrix

Test ID	Test	Expected Result	Status
TC01	Backend startup	API available	Passed
TC02	Frontend availability	UI available	Passed
TC03	Machine retrieval	Valid machine JSON	Passed
TC04	Fault injection	Alert and risk generated	Passed

Automated Test Summary:

Total Tests = 4

Passed = 4

Failed = 0

Execution Time = 2.19 seconds.

CHAPTER 26: COURSE EXPERIENCE

26.1 Overview of the Course Experience

The PredictSense project provided a practical learning experience that connected academic concepts with an end-to-end software implementation. The work required the application of machine learning, data analysis, backend development, database management, explainable AI, retrieval-augmented generation, agentic workflows and frontend engineering within one integrated system.

26.2 Technical Skills Applied

Area	Experience / Application
Data Analysis	Dataset inspection, statistical analysis, missing-value verification, failure-distribution analysis and exploratory analysis.

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Machine Learning	Isolation Forest for anomaly detection and XGBoost for multi-label failure-mode prediction.
Feature Engineering	Temperature difference, angular velocity, mechanical spindle power and wear-torque features.
Explainable AI	SHAP-based feature attribution to understand the contribution of sensor variables.
Backend Development	FastAPI REST services for telemetry, predictions, alerts, maintenance and agent execution.
Database Engineering	PostgreSQL/Supabase persistence for machines, telemetry, predictions, alerts and maintenance actions.
Generative/Agentic AI	LangGraph orchestration and RAG retrieval for maintenance-oriented diagnostic assistance.
Frontend Development	React and Vite components for fleet monitoring, machine inspection, alerts and knowledge-base interaction.
Testing	API verification, automated pytest execution, database verification and controlled fault injection.

26.3 Learning from End-to-End Integration

A major learning outcome was understanding that an industrial AI solution cannot be treated as an isolated machine-learning model. The project required connecting data ingestion, validation, feature engineering, model inference, explainability, risk scoring, knowledge retrieval, recommendations, database persistence and user interaction. This provided experience in designing interfaces between different technical layers.

26.4 Practical Problem-Solving Experience

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The project provided experience in handling practical engineering issues such as highly imbalanced failure classes, sparse Random Failure examples, API integration, persistent database storage, automated testing and controlled fault simulation. The AI4I dataset contains only nine Random Failure positive records, which demonstrated why precision, recall, F1 score, ROC-AUC and PR-AUC are important instead of relying only on accuracy.

26.5 Experience with Explainable and Agentic AI

The integration of SHAP, LangGraph and RAG provided practical exposure to modern AI system design. SHAP was used to connect model predictions with feature-level evidence, while LangGraph organized diagnostic steps and RAG connected the reasoning process to maintenance SOP information. This demonstrated the difference between predicting a failure and supporting an actual maintenance decision.

26.6 Software Engineering and Testing Experience

The project strengthened software-engineering practices through modular backend services, REST API design, database persistence, frontend integration and automated testing. The supplied project validation recorded four automated tests with all four passing in 2.19 seconds, together with successful backend and frontend HTTP verification. Fault injection was also used to verify that an abnormal telemetry condition could propagate through the system and generate an elevated risk and priority alert.

26.7 Professional and Academic Learning

The project improved the ability to convert a problem statement into requirements, architecture, implementation, testing and technical documentation. It also developed experience in communicating technical results through tables, system workflows, dashboards, model metrics, code listings and case-study analysis.

26.8 Overall Course Experience

Overall, the course/project experience provided a complete view of applied artificial intelligence and full-stack engineering. Instead of learning each technology independently, the project demonstrated how data science, machine learning, backend systems, databases, explainable AI, agentic AI and frontend development can work together to solve a realistic industrial decision-support problem.

26.9 Key Learning Outcomes

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No.	Learning Outcome
1	Understand the complete predictive-maintenance lifecycle.
2	Build and evaluate anomaly and failure-diagnosis models.
3	Engineer domain-relevant features from raw telemetry.
4	Interpret model predictions using SHAP.
5	Design REST APIs and persistent database workflows.
6	Integrate LangGraph and RAG into a decision-support pipeline.
7	Build an interactive monitoring dashboard.
8	Validate an integrated system using automated tests and fault injection.
9	Understand limitations and responsible deployment requirements.